

Technical Readiness Template

Coordinating PI:	
Project Title:	

Code Links

Provide the link(s) to your relevant GitHub repositories in URL format. If your repository is currently private, please add the following users as collaborators: the Kempner Institute (GitHub username: kempner-compute).

1 .	e.g., https://github.com/github/docs
2 .	
3 .	
4 .	

Frameworks, Libraries, and Software Environment

Please specify the GPU-accelerated framework you intend to use (e.g. PyTorch, TensorFlow) along with any other AI/ML frameworks, software dependencies, or libraries that are required for your project. If the workflows are containerized, please list what containerization tools you are using. (Maximum of 200 words)

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External APIs and AI Services

Specify if your project would utilize any external APIs or AI services (e.g., OpenAI, Anthropic, or Hugging Face) and explain their specifications (e.g., API call frequency) and function within your workflow.

GPU Type

Specify the type of GPUs required (e.g., A100, H100 or mixed) and whether your workload depends on specific GPU features (e.g., low precision, such as FP8, or large VRAM).

Data Requirements

Describe the datasets you will be using. Please include their size and current storage location (e.g., AWS, on-premises, local devices). We recommend Globus (read more [here](#)) for faster file transfer. **Please note that only data that can be made publicly available and rated data [security level 1 or 2](#) is permitted.**

Storage Needs

How much storage will your project require for input data, models, container images, and other training artifacts? Do you anticipate that your model will generate large output files and checkpoints, if so, please estimate their size. Please round up to the nearest terabyte.

Technical Expertise

Briefly describe the project team's technical expertise in using GPUs, distributed systems, or large-scale AI/ML training.

Deliverables

List expected deliverables following the conclusion of the award (e.g., codebase, notebooks, datasets, models, weights, etc.).

Scalability Testing

Please report system usage statistics for multi-GPU jobs when running the proposed workflow. We require testing to include a minimum of a *quarter of the requested GPUs* and ask that GPU type, GPU count, average peak memory per GPU, and average GPU utilization are reported. We recommend using [KempnerPulse](#), a GPU monitoring system that integrates into any workflow. Please submit the results using the export function and [export all](#) of the usage statistics. We also recommend sharing a SLURM job submission script configuration that has been tested on a high-performance computing cluster (although, not required). If you currently do not have access to an HPC cluster, there are resources available through the Harvard ecosystem such as [kempner_requeue](#), [public_requeue partitions \(FAS\)](#), and the [Research Computing Core](#) (Harvard Medical School) that allow for testing.

We have provided an example of a report here: [Sample Training Report](#) that was authored by a Kempner Affiliate member as a reference point.

Num. Nodes	Total Num. GPUs/DP Global Rank	Time to 1000 steps - Time to 500 steps (mins)	Peak Memory per GPU (GB)	Runtime Speedup from 1 Node
1	4			N/A
4	16			x__